3D HERITAGE ENGINEER

Maxime Lafleur

Engineer · Photogrammetrist · Lasergrammetrist · Developer

Passionate about cultural heritage and cutting-edge 3D reconstruction technologies, I initially built my expertise around drone geomatics — from hardware and software development to applied research — gaining full autonomy over the entire process, from study and prototyping to product development (2014–2022).

More recently (2022–2026), my work in the cultural heritage sector immersed me in high-fidelity 3D capture and reconstruction. The technical challenges and the blending of diverse technologies meant mastering specialized skills and methodologies, pushing existing solutions to their limits, and developing automation to eliminate repetitive tasks.

With 3D technology evolving rapidly and opening up new possibilities, **I'm eager to keep exploring this field**, learn from others and draw on new approaches, as part of a creative team — bringing my 12 years of curiosity and varied experience.

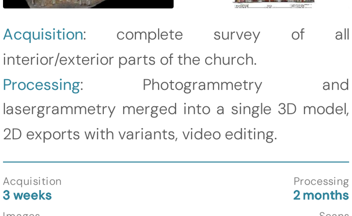
Looking forward to connecting and discussing projects, opportunities and experiences!



GEOKALI · FIELD PROJECTS

Key Heritage & Civil Engineering projects at Geokali

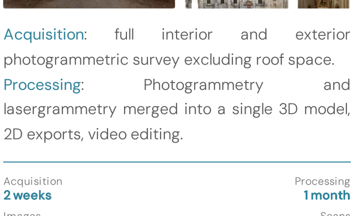
01 Château de Chenonceau & dépendance...



Acquisition: Full exterior / interior survey — château, Tour des Marques and all outbuildings.
Processing: Photogrammetry, lasergrammetry, 2D exports, video editing.

- Acquisition: 3 months / 2 years
- Processing: 9 months
- Images: ~265 000
- Scans: ~5 500
- 3D Models: ~150 · 939M △
- Texture resolution: 1 mm
- Project size: 16 TB
- 2D AutoCAD drawings: 173 .dwg · ~700 ortho

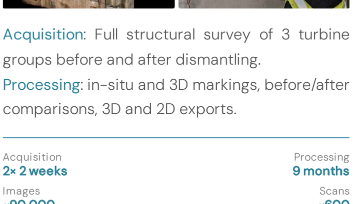
02 Cathédrale de Nevers



Acquisition: Hybrid laser/photo survey interior/exterior based on client needs and accessible laser geometry.
Processing: Photogrammetry and segmented lasergrammetry, 2D exports and video editing.

- Acquisition: 1 month
- Processing: 2 months
- Images: ~46 000
- Scans: ~1 830
- 3D Models: ~30 · 237M △
- Texture resolution: 1 mm
- 2D AutoCAD drawings: 57 .dwg · ~200 ortho

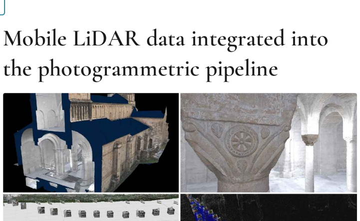
03 Basilique de Fourvière — Lyon



Acquisition: Full exterior laser survey and photo of upper parts (exterior and interior).
Processing: Photogrammetry and segmented lasergrammetry, 2D exports, video editing.

- Acquisition: 2 weeks
- Processing: 1 month
- Images: ~38 000
- Scans: ~400
- 3D Models: ~20 · 189M △
- Texture resolution: 1 mm
- 2D AutoCAD drawings: 57 .dwg · ~200 ortho

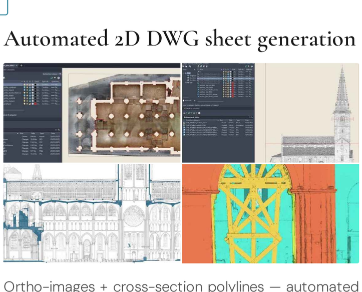
04 Église Notre-Dame — Dijon



Acquisition: complete survey of all interior/exterior parts of the church.
Processing: Photogrammetry and lasergrammetry merged into a single 3D model, 2D exports with variants, video editing.

- Acquisition: 3 weeks
- Processing: 2 months
- Images: ~54 000
- Scans: ~1 100
- 3D Models: ~2 · 140M △
- Texture resolution: 1 mm
- 2D AutoCAD drawings: 40 .dwg · ~160 ortho

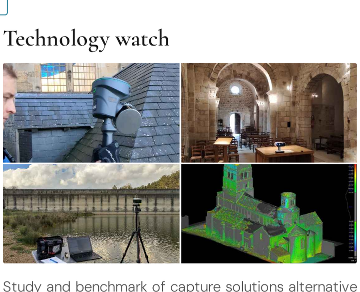
05 Église Saint-Philibert — Dijon



Acquisition: full interior and exterior photogrammetric survey excluding roof space.
Processing: Photogrammetry and lasergrammetry merged into a single 3D model, 2D exports, video editing.

- Acquisition: 2 weeks
- Processing: 1 month
- Images: ~30 000
- Scans: ~250
- 3D Models: ~1 · 15M △
- Texture resolution: 1 mm
- 2D AutoCAD drawings: 4 .dwg · ~40 ortho

06 Civil Engineering Structure



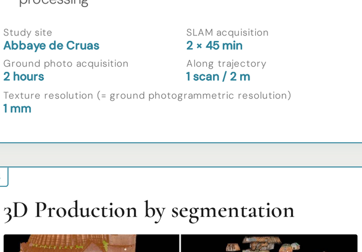
Acquisition: Full structural survey of 3 turbine groups before and after dismantling.
Processing: in-situ and 3D markings, before/after comparisons, 3D and 2D exports.

- Acquisition: 2× 2 weeks
- Processing: 9 months
- Images: ~90 000
- Scans: ~600
- 3D Models: 3 · 50M △
- Texture resolution: 0.35 mm
- Project size: 8 TB
- Defects (Cyclone 3DR): ~900 m
- 2D AutoCAD drawings: 30 .dwg · ~120 ortho

INNOVATION & DEVELOPMENT

Key Heritage R&D projects at Geokali

01 Mobile LiDAR data integrated into the photogrammetric pipeline



Integration of mobile scanner / LiDAR data into the standard photogrammetric workflow. Interior with mobile scanner, exterior with mobile scanner supported by photogrammetric capture.

Benefits:

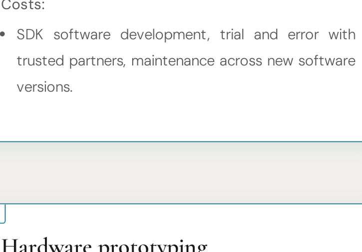
- Maintain a standard 3D production workflow only, replacing static scans with virtual positions from SLAM
- Panoramic photo contribution from the SLAM scanner, equivalent to static panoramas

Costs:

- Georeferencing precision, drift analysis, strict GCP setup — strong mastery of SLAM solutions and their limits
- Risk of geometric fidelity failure (drift, double surfaces) with no correction capacity during processing

- Study site: Abbaye de Cruas
- SLAM acquisition: 2 × 45 min
- Ground photo acquisition: 2 hours
- Along trajectory: 1 scan / 2 m
- Texture resolution (≈ ground photogrammetric resolution): 1 mm

02 Automated 2D DWG sheet generation



Ortho-images + cross-section polylines — automated generation of DWG sheets (plans / sections / elevations).

Benefits:

- Automate DWG extraction generation (plans / sections / elevations)
- Provide an internal operator with an export selection tool for resolutions and client-subscribed options (depth views, contours, confidence map, pure 3D view)
- Eliminate any human error in the processing chain
- Avoid chained operations during extractions
- Systematize deliverables and lock nomenclature, extraction themes and renders
- Respond quickly to client requests — extraction productivity multiplied by ×30 per .dwg sheet

Costs:

- SDK software development, trial and error with trusted partners, maintenance across new software versions.

03 Technology watch

Study and benchmark of capture solutions alternative or complementary to static scanners: mobile scanners, mobile SLAM, high-resolution photogrammetry — to optimize acquisition time and adapt methods to site geometry.

04 3D Production by segmentation

Benefits:

- Reducing 3D processing time and complexity
- Splitting work between 3D operators and computers
- Ability to reprocess a specific area following potential changes
- Adapting detail level and texture quality per zone
- Offering clients optimized acquisition costs

Costs:

- Careful cleaning of all laser scans, filtering of images / laser stations, 3D model cleaning, strict file naming conventions.

05 Hardware prototyping

Optimizing acquisition systems by adding purpose-built accessories adapted to field constraints.

Identified needs:

- Accessing confined spaces
- Securing the system for specific use cases
- Simplifying daily operations (heavy handheld equipment)

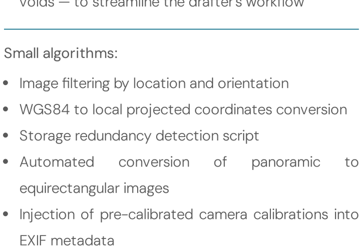
Adopted solutions:

- Laser scanner trolley (productivity gain)
- 3D camera gimbal for repetitive facades or confined spaces (DJI Ronin)
- Steadycam vest and belt for heavy equipment (Flycam Galaxy)

Custom solutions:

- Low-profile tripod for terrestrial scanner (daily use)
- Camera addon fixed to terrestrial laser scanner
- Custom attachment systems for gimbals

06 Software prototyping



Pushing daily-use software to its limits to improve working comfort, deliverable quality and processing time, and to develop key algorithms for acquisition / processing methodologies.

Automated script development for:

- AutoCAD / Photoshop / Leica Cyclone 3DR / Agisoft Metashape / RealityScan

Developed tools:

- Multi-scale analysis tool for alignment discontinuities from RealityScan photogrammetric processing
- Automated image QC tool (motion/focal blur, exposure) with renaming, copying and raw data backup
- Ortho-image generation with various renders: depth, contours, Xray, 3D confidence, geometric voids — to streamline the drafter's workflow

Small algorithms:

- Image filtering by location and orientation
- WGS84 to local projected coordinates conversion
- Storage redundancy detection script
- Automated conversion of panoramic to equirectangular images
- Injection of pre-calibrated camera calibrations into EXIF metadata
- And other small useful developments on a project-by-project basis

PROFESSIONAL EXPERIENCE

- **Cyclope3D** — JUL. 2026 → PRESENT · 3 MO
- **Geokali** — 2022 → JUNE 2026 · 3 YRS 9 MO
- **Microdrones™ / GeoCue** — 2016 → 2022 · 6 YRS 5 MO
- **CNRS** — 2021 → 2022 · 1 YR 6 MO
- **Delair** — 2016 → 2018 · 2 YRS
- **InventUAV** — 2015 → 2016 · 1 YR
- **Vision du Ciel Intl** — 2014 → 2015 · 1 YR
- **Groupe SII · Nexter** — 2013 → 2014 · 1 YR

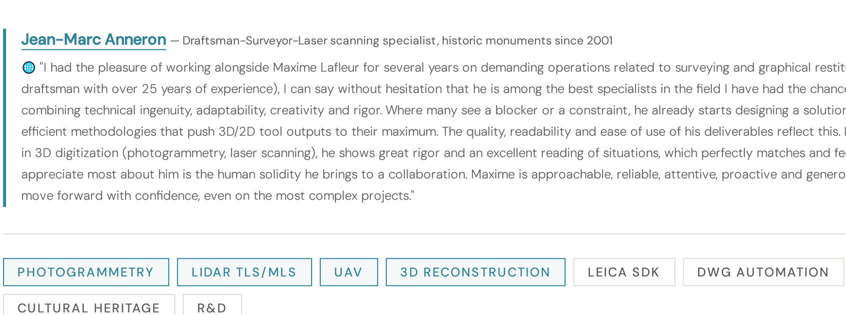
Cyclope3D Software Publisher

Cyclope3D · Freelance · Auvergne-Rhône-Alpes, France — 3 mo, Jul. 2026 → present

Design and development of **Cyclope3D**, a tool for extracting 2D drawings from 3D data (meshes, point clouds, 3DGS), built for drafters, architects, and laser scanning and photogrammetry surveyors. The tool draws on several years of client feedback and analysis of the deliverables on the market.

- Opening and analysis of 3D deliverables, to get to grips with the survey and check it against the specifications
- Automatic 2D drawing sheet generation, with no manual handling: georeferenced orthoimages, DXF/DWG drawings and PDF sheets
- Several representations per view (texture, geometry, depth, solid areas, X-ray), overlaid in the drawing
- Guided naming of folders, files and layers, derived from the project and view names
- Customizable deliverables in the company's colors: orthoimage background, logo

→ cyclope3d.fr



SOFTWARE DEVELOPMENT · 2D DRAWING EXTRACTION · DXF / DWG · ORTHOIMAGES · POINT CLOUDS · 3DGS · CULTURAL HERITAGE

Photogrammetry / Lasergrammetry Engineer

Geokali · Full-time · Valence, France — 3 yrs, Oct. 2022 → June 2026

3D data acquisition and processing on **cultural heritage and civil engineering** sites, using a full sensor fleet: UAV drones, terrestrial laser scanners (TLS), mobile laser scanners (MLS), DSC cameras and structured light lasers.

→ See Heritage & Civil Engineering projects · See R&D projects

> **Jean-Marc Anneron** — Draftsman-Surveyor-Laser scanning specialist, historic monuments since 2001
> "I had the pleasure of working alongside Maxime Lafleur for several years on demanding operations related to surveying and graphical restitution of historic monuments. Looking back (and as a Heritage draftsman with over 25 years of experience), I can say without hesitation that he is among the best specialists in the field I have had the chance to collaborate with. Maxime has an extremely rare profile, combining technical ingenuity, adaptability, creativity and rigor. Where many see a blocker or a constraint, he already starts designing a solution: he codes, develops routines, automates complex tasks and creates efficient methodologies that push 3D/2D tool outputs to their maximum. The quality, readability and ease of use of his deliverables reflect this. In the field, his mastery of capture operations is remarkable. Whether in 3D digitization (photogrammetry, laser scanning), he shows great rigor and an excellent reading of situations, which perfectly matches and feeds Geokali's expertise. But beyond the skills, what I probably appreciate most about him is the human solidity he brings to a collaboration. Maxime is approachable, reliable, attentive, proactive and generously involved. When working with him, you immediately feel you can move forward with confidence, even on the most complex projects."

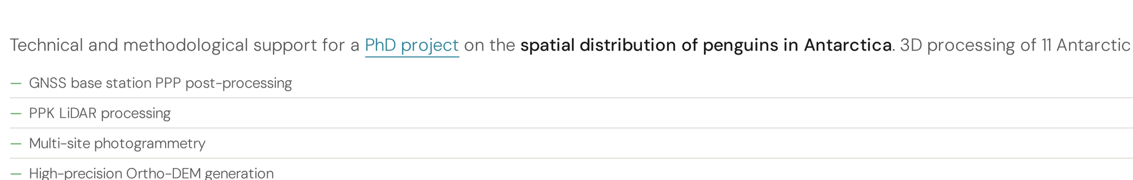
PHOTOGRAMMETRY · LIDAR TLS/MLS · UAV · 3D RECONSTRUCTION · LEICA SDK · DWG AUTOMATION · POINT CLOUDS · HARDWARE PROTOTYPING · CULTURAL HERITAGE · R&D

UAV LiDAR Engineer

Microdrones™ / GeoCue · R&D · Toulouse — 6 yrs, May 2016 → Sept. 2022

Within the R&D department of Microdrones / GeoCue, responsible for developing **new UAV-embedded LiDAR prototypes**, designing high-precision LiDAR targets, and QC of new processing software.

- Operational setup of Microdrones France subsidiary (office design, construction follow-up, HR)
- Hardware design of UAV LiDAR systems
- Field testing and data QC
- Hardware and software benchmarks
- Development of an automated tree counting tool
- Automated powerline classification
- Parachute system development and EASA certification approval



> **Vincent Eucat** — Industrial CAD Draftsman at mdGroup
> "I had the opportunity to work alongside Maxime Lafleur during my time at Microdrones, and I can wholeheartedly vouch for his excellent skills as an engineer. As part of our drone data acquisition missions for various vineyard estates in France, Maxime was a true technical pillar. His analytical mindset and great resourcefulness allowed him to efficiently diagnose and solve complex hardware and software issues, often under demanding field conditions. Whether it was last-minute adjustments to the equipment or optimizing flight operations, his ingenuity always ensured the continuity and quality of our operations. I highly recommend him to any organization looking for a reliable and talented field engineer."

> **Erwan Gavelle** — Founder of geogrid
> "I had the pleasure of working with Maxime on the development of drone payloads. He was in charge of the electronics and project management while I worked on the mechanics. It was my first job in this field, but he trusted me, taught me and challenged me throughout the projects. I particularly appreciated his skills, his involvement and his friendliness in driving the projects to success. He also managed a large part of the organization and setup of our offices in France. I highly recommend Maxime for his professionalism, his good mood and his ability to support the people he works with."

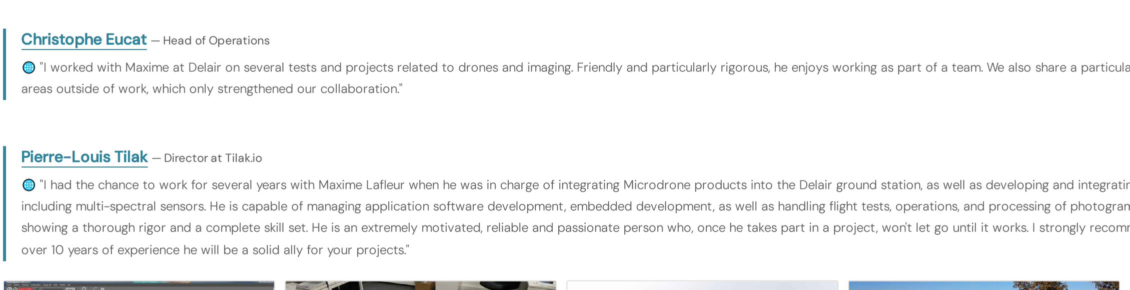
UAV LIDAR R&D · HARDWARE DESIGN · EASA CERTIFICATION · POWERLINE CLASSIFICATION · TREE COUNTING · FIELD TESTING

3D / GIS Support for Antarctic PhD Project

CNRS · Civic Service · Remote — 1.5 yrs, Feb. 2021 → Jul. 2022

Technical and methodological support for a PhD project on the **spatial distribution of penguins in Antarctica**. 3D processing of 11 Antarctic sites.

- GNSS base station PPP post-processing
- PPK LiDAR processing
- Multi-site photogrammetry
- High-precision Ortho-DEM generation



ANTARCTICA · PPK LIDAR · GNSS PPP · PHOTOGRAMMETRY · ORTHO-DEM · GIS

UAV Engineer Payload Integration & Control/Command

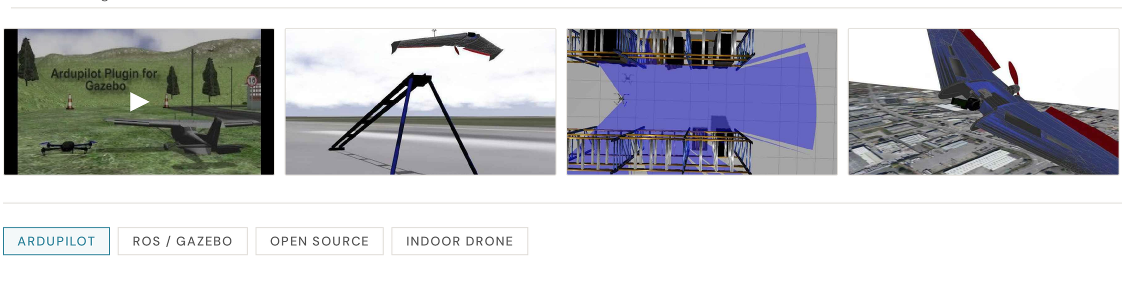
Delair · Full-time · Toulouse — 2 yrs, May 2016 → May 2018

Within Delair-Tech's R&D department, under Microdrones' strategic direction, responsible for:

- Control/command system development for Microdrones UAVs
- Integration of Delair-Tech payloads (multispectral, optronic, photogrammetry) on Microdrones md4-1000
- Viticulture research project in partnership with Naïo Technologies (vineyard row mapping with PPK)
- Other prototyping projects

> **Christophe Eucat** — Head of Operations
> "I worked with Maxime at Delair on several tests and projects related to drones and imaging. Friendly and particularly rigorous, he enjoys working as part of a team. We also share a particular interest in several areas outside of work, which only strengthened our collaboration."

> **Pierre-Louis Tilak** — Director at Tilak.io
> "I had the chance to work for several years with Maxime Lafleur when he was in charge of integrating Microdrone products into the Delair ground station, as well as developing and integrating several payloads, including multi-spectral sensors. He is capable of managing application software development, embedded development, as well as handling flight tests, operations, and processing of photogrammetry data, showing a thorough rigor and a complete skill set. He is an extremely motivated, reliable and passionate person who, once he takes part in a project, won't let go until it works. I strongly recommend him, as with over 10 years of experience he will be a solid ally for your projects."



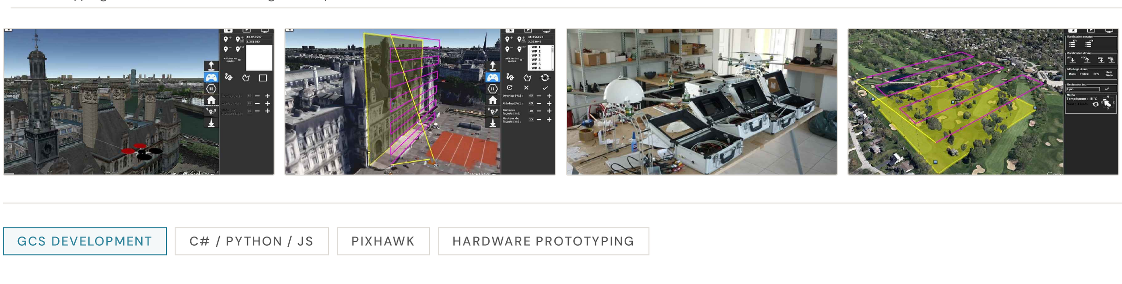
PAYLOAD INTEGRATION · FIXED-WING UAV · EMBEDDED SYSTEMS

Co-Founder

InventUAV · Open-Source Startup · France — 1 yr, May 2015 → May 2016

Co-founder of an open-source UAV startup developing an **autonomous indoor drone** for warehouse management and inventory.

- ArduPlane plugin for Gazebo / ROS
- ArduCopter debugging
- Visual recognition for indoor missions



ARDUPILOT · ROS / GAZEBO · OPEN SOURCE · INDOOR DRONE

UAV Apps Engineer (GCS / GUI)

Vision du Ciel Intl · Orange — 1 yr, Jul. 2014 → Jun. 2015

Development of **UAV mission control** software solutions on Windows: mission planning, control, supervision and reporting.

- GUI development in C#, JS, Python
- Embedded development in C++
- Open-source and industrial protocol integration
- Prototyping: UAVs, radio controllers, gimbals, parachutes

GCS DEVELOPMENT · C# / PYTHON / JS · PIXHAWK · HARDWARE PROTOTYPING

System Architect Nexter Systems

Groupe SII · Satory — 1 yr, Oct. 2013 → Jul. 2014

Writing, analysis and capitalisation of system requirements for Nexter. Architecture in IBM DOORS, document traceability, automation script development, object-oriented C++ modelling.

SYSTEM ARCHITECTURE · IBM DOORS · C++ · REQUIREMENTS ANALYSIS

More information available on request

TECHNICAL EXPERTISE

Skills & tools (non exhaustive)

Acquisition & Field	Software	Development
Hardware prototyping (tripod, steadycam, gimbal)	RealityCapture / RealityScan / PostShot / Metashape	Multi-source 3D reconstruction (photo + TLS + MLS + structured light)
UAV / drone (multirotor, fixed-wing)	Claude AI	Automated tools (plan extraction, QC, AI denoising)
Camera calibration (intrinsic & extrinsic)	Point cloud processing & cleaning	JavaScript
TLS — Terrestrial Laser Scanning	homemade apps	AI experiments (photogrammetry, Android apps)
Aerial & terrestrial photogrammetry	Z+F Laser Control Office / Scantra	C++ / C#
ROV: underwater drones	Unreal Engine 5 / Cinema 4D	System architecture & requirements (UML, DOORS)
MLS — Mobile Laser Scanning and Lidars	GNSS processing (PPK/PPP)	ArduPilot / APM
Field quality control (QA/QC)	Solidworks & 3D printing	Processing pipeline (farm processing)
GNSS / PPK / PPP processing	Leica Cyclone3DR	Scanner hardware benchmarking
Structured light scanner	Postshot & 3DGS	GUI development (Windows, GCS mission control)
	GIS (QGIS)	Electronic prototyping
	CloudCompare	DWG export / automated 2D drawings
	AutoCAD	Payload integration
	Blender	ROS / Gazebo
		EXIF scripting
		Python

OFF THE CLOCK

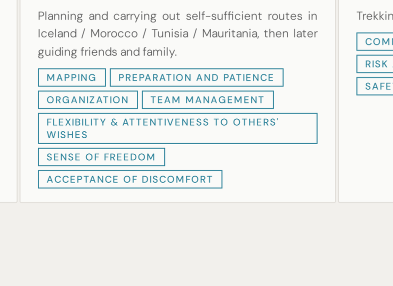
Personal experiences and associated skills

Photography passion

Film development & scanning, night photography, astrophotography

CURIOSITY · PATIENCE · RIGOR

Azores Project: discovering the world of sailing



Buying the sailboat, learning sailing & navigation principles, onboard medical training, test trips, equipment/safety audit, and the round-trip voyage.

SELF-TEACHING · RIGOROUS PREPARATION · SELFLESSNESS · MENTAL & PHYSICAL RESILIENCE · FREEDOM · ACCEPTANCE OF DISCOMFORT

Archiving / passing on knowledge

Photogrammetry of sites/objects, family tree research and photo archiving, historical research

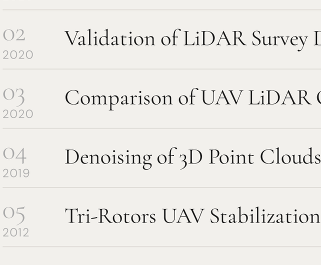
CURIOSITY · PATIENCE · RIGOR · JOY OF SHARING KNOWLEDGE

Support

Managing a non-profit website, opening a CAF mountain hut, sailboat delivery, mechanical help, 3D design & printing, Android apps

SELF-TEACHING · PATIENCE · JOY OF SHARING KNOWLEDGE

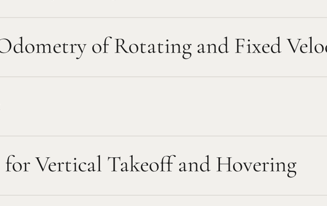
Highlands Project: cycling



Scotland Highlands project: route mapping, wild camping and self-sufficiency

PREPARATION · PHYSICAL & MENTAL MANAGEMENT · FREEDOM

Desert Discovery



Planning and carrying out self-sufficient routes in Iceland / Morocco / Tunisia / Mauritania, then later guiding friends and family.

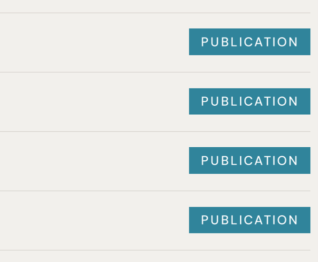
MAPPING · PREPARATION AND PATIENCE · ORGANIZATION · TEAM MANAGEMENT · FLEXIBILITY & ATTENTIVENESS TO OTHERS' WISHES · SENSE OF FREEDOM · ACCEPTANCE OF DISCOMFORT

Active pursuits

Trekking, mountaineering, competition, climbing

COMMITMENT · DEDICATION · RISK ACCEPTANCE · MENTAL MANAGEMENT · SAFETY · SENSE OF FREEDOM

Aviation passion



Paragliding training and practice, discovering the physics of gliders/planes/helicopters, simulators.

CURIOSITY · MENTAL MANAGEMENT · RIGOR · PATIENCE

RESEARCH & INNOVATION

Publications & Patents

#	Year	Title	Type
01	2020	Patent : 3D Accuracy Star + Catenary Lines	PATENT
02	2020	Validation of LiDAR Survey Data by Comparison of Several Uncertainty Models	PUBLICATION
03	2020	Comparison of UAV LiDAR Odometry of Rotating and Fixed Velodyne Platforms	PUBLICATION
04	2019	Denoising of 3D Point Clouds	PUBLICATION
05	2012	Tri-Rotors UAV Stabilization for Vertical Takeoff and Hovering	PUBLICATION

EDUCATION

Education & International exchanges

- **EU** — European Drone Certificate — Open A1/A2/A3 and Specific CATS — 2026
- ECE Paris — Master of Engineering (MEng) Electrical & Electronics Engineering — 2008–2013
- **US** — University of California, Irvine — Exchange Program Global Management & Manufacturing — 2012
- **CA** — Concordia University — Semester Exchange Engineering & Computer Science — 2011

CONTACT

Let's connect !

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- EMAIL: maxime.lafleur@outlook.fr
- MOBILITY: Rhône-Alpes → France → Europe → World
- LANGUAGES: French (native) · English (Full Professional)

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